LAB PROGRAM NO:1

Programiz

C Online Compiler

main.c

Share

Run

```c
#include<stdio.h>
#include<unistd.h>
int main()
{
    printf("Process id:%d\n",getpid());
    printf("Parent process id:%d\n",getpid());
    return 0;
}
```

Output

```
Process id:3011
Parent process id:3011
```

LAB:2

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main.c

```c
#include <stdio.h>
#include <stdlib.h>
int main()
{
FILE *fptr1, *fptr2; char
filename[100], c;
printf("Enter the filename to open for reading \n");
scanf("%s", filename);
fptr1 = fopen(filename, "r");if
(fptr1 == NULL)
{
printf("Cannot open file %s \n", filename);
exit(0);
}
printf("Enter the filename to open for writing \n");
scanf("%s", filename);
fptr2 = fopen(filename, "w");if
(fptr2 == NULL)
{
printf("Cannot open file %s \n", filename);
exit(0);
}
c = fgetc(fptr1);
while (c != EOF)
{
fputc(c, fptr2); c
```

```
printf("Cannot open file %s \n", filename);
exit(0);
}
c = fgetc(fptr1);
while (c != EOF)
{
fputc(c, fptr2); c
= fgetc(fptr1);
}
printf("\nContents copied to %s", filename);
fclose(fptr1);
fclose(fptr2);
return 0;
}
```

Output Clear

```
Enter the filename to open for reading
Hello World!
Welcome to C programming.
File handling is easy.
Cannot open file Hello


=== Code Execution Successful ===
```

SPONSO
ESTATES

LAB:3

Programiz

C Online Compiler

Smoltek's CNF-MIM techr

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main.c

Share

Run

```c
#include <stdio.h>
int main()
{
int A[100][4];
int i, j, n, total = 0, index, temp;float
avg_wt, avg_tat;
printf("Enter number of process: ");
scanf("%d", &n);
printf("Enter Burst Time:\n");for (i
= 0; i < n; i++) {
printf("P%d: ", i + 1);
scanf("%d", &A[i][1]);
A[i][0] = i + 1;
}
for (i = 0; i < n; i++) {
    index = i;
for (j = i + 1; j < n; j++)
if (A[j][1] < A[index][1])
index = j;
temp = A[i][1]; A[i][1] =
A[index][1];A[index][1]
= temp;
temp = A[i][0]; A[i][0] =
A[index][0];A[index][0]
= temp;
}
```

```
A[0][2] = 0;
for (i = 1; i < n; i++) {
A[i][2] = 0;
for (j = 0; j < i; j++)
A[i][2] += A[j][1];
total += A[i][2];
}
avg_wt = (float)total / n;
total = 0;
printf("P BT WT TAT\n");
for (i = 0; i < n; i++) {
A[i][3] = A[i][1] + A[i][2];
total += A[i][3];
printf("P%d %d %d %d\n", A[i][0],A[i][1],
A[i][2], A[i][3]);
}
avg_tat = (float)total / n;
printf("Average Waiting Time= %f", avg_wt);
printf("\nAverageTurnaround Time= %f", avg_tat);
}
```

Output

```
Enter number of process: 4
Enter Burst Time:
P1: 4
P2: 6
P3: 7
P4: 5
P BT WT TAT
P1 4 0 4
P4 5 4 9
P2 6 9 15
P3 7 15 22
Average Waiting Time= 7.000000
AverageTurnaround Time= 12.500000

=== Code Execution Successful ===
```

LAB:5

Programiz

C Online Compiler

main.c

```c
#include <stdio.h>
int main() {
int n, i, j;
printf("Enter number of processes: ");
scanf("%d", &n);
int bt[n], wt[n], tat[n], priority[n], p[n];
for(i = 0; i < n; i++) {
    p[i] = i + 1;
    printf("\nEnter Burst Time and Priority for Process P%d: ", p[i]
        );
    scanf("%d %d", &bt[i], &priority[i]);
}
for(i = 0; i < n - 1; i++) {
    for(j = i + 1; j < n; j++) {
        if(priority[i] > priority[
            int temp = priority[i];
            priority[i] = priority[j];
            priority[j] = temp;
            temp = bt[i];
            bt[i] = bt[j];
            bt[j] = temp;
            temp = p[i];
            p[i] = p[j];
            p[j] = temp;
        }
    }
```

```
                bt[j] = temp;
                temp = p[i];
                p[i] = p[j];
                p[j] = temp;
            }
        }
    }
    wt[0] = 0;
    for(i = 1; i < n; i++) {
        wt[i] = wt[i-1] + bt[i-1];
    }
    for(i = 0; i < n; i++) {
        tat[i] = wt[i] + bt[i];
    }
    printf("\nProcess\tBurst Time\tPriority\tWaiting Time\tTurnaround
        Time\n");
    for(i = 0; i < n; i++) {
        printf("P%d\t\t%d\t\t%d\t\t%d\t\t%d\n",
                p[i], bt[i], priority[i], wt[i], tat[i]);
    }
    return 0;
    }
```

Output Clear

```
Enter number of processes: Enter number of processes: 4



Enter Burst Time and Priority for Process P1: 5 2

Enter Burst Time and Priority for Process P2: 3 1

Enter Burst Time and Priority for Process P3: 8 4

Enter Burst Time and Priority for Process P4: 6 3

Process Burst Time  Priority    Waiting Time    Turnaround Time


=== Code Execution Successful ===
```

LAB:4

Programiz

C Online Compiler

Programiz PRO

Never Learn

main.c

Share

Run

```c
#include<stdio.h>
int main()
{
int bt[20],p[20],wt[20],tat[20],i,j,n,total=0,pos,temp;float
avg_wt,avg_tat;
printf("Enter number of process:");
scanf("%d",&n);
printf("nEnter Burst Time:\n");for(i=0;i<n;i++)
{
printf("p%d:",i+1);
scanf("%d",&bt[i]);
p[i]=i+1;
}
for(i=0;i<n;i++){
pos=i;
for(j=i+1;j<n;j++)
{
if(bt[j]<bt[pos])
pos=j;
}
temp=bt[i];
bt[i]=bt[pos];
bt[pos]=temp;
temp=p[i];
p[i]=p[pos];
p[pos]=temp;
```

```c
p[pos]=temp;
}
wt[0]=0;
for(i=1;i<n;i++)
{
wt[i]=0;
for(j=0;j<i;j++)
wt[i]+=bt[j];
total+=wt[i];
}
avg_wt=(float)total/n;
total=0;
printf("nProcesst Burst Time tWaiting TimetTurnaround Time\n");
for(i=0;i<n;i++)
{
tat[i]=bt[i]+wt[i];
total+=tat[i];
printf("np%dtt %dtt %dttt%d",p[i],bt[i],wt[i],tat[i]);
}
avg_tat=(float)total/n;
printf("nnAverage Waiting Time=%f",avg_wt);
printf("nAverage Turnaround Time=%fn",avg_tat);
}
```

Output Clear

```
Enter number of process:3
nEnter Burst Time:
p1:3
p2:2
p3:6
nProcesst Burst Time tWaiting TimetTurnaround Time
np2tt 2tt 0ttt2np1tt 3tt 2ttt5np3tt 6tt 5ttt11nnAverage Waiting Time=2
    .333333nAverage Turnaround Time=6.000000n

=== Code Execution Successful ===
```